Preserving PET Quantification in Longitudinal Whole-Body Registration

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Abstract. Longitudinal PSMA PET/CT registration has to align whole-body scans that differ in field of view, positioning , disease burden and bowel and bladder filling, while preserving the PET-derived biomarkers that the scans are acquired to measure. Accuracy and preservation pull against each other: a deformation that maximises organ overlap is free to compress or expand a lesion, changing exactly the quantity a clinician reads. We adapt the diffeomorphic, coarse-to-fine LapIRN architecture to this task for the Learn2Reg 2026 PSMAReg challenge, and add two anatomically motivated constraints to its objective. The first is a set of biomarker terms that pin the metabolic tumour volume and total lesion glycolysis under the estimated transformation, including a Jacobian-based surrogate that supplies gradients throughout the lesion interior rather than only at its boundary. The second is a bone-rigidity penalty formulated as a per-structure rigid fit: classical derivative-based rigidity terms read a finite-difference stencil that, on ribs and cortical bone at whole-body resolution, is dominated by surrounding soft tissue, whereas a closed-form rigid fit per skeletal label reads nothing outside it. The same objective is descended per case at test time by instance optimisation. On the challenge validation cases the submitted container reaches 81.0% DSC and 6.17 mm HD95 at MTV and TLG errors of 3.3%, against 73.3%, 7.09 mm and 4.5% and 8.7% for the strongest baseline; released from the 90 s per pair the container must meet, the same method reaches 88.3% DSC, 4.80 mm HD95 and 1.4% and 2.1% biomarker error.

Keywords: Deformable image registration · PSMA PET/CT · Longitudinal imaging · Biomarker preservation.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [6]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans afterwards; accurate baseline-to-follow-up registration can support comparison of disease burden, localisation of uptake changes, assessment of lesion progression or response, dose accumulation, and quantitative longitudinal analysis [14]. The challenge organisers identify several factors that make establishing this correspondence by deformable image registration difficult in routine whole-body data [14]: baseline and follow-up scans differ substantially in field of view, patient positioning, respiratory state, disease burden, bladder and bowel filling, and body weight; CT and PET encode complementary but appearance-dissimilar information; and the registration must preserve the quantitative PET signal used for response assessment while remaining physically plausible in structures such as the skeleton. The Learn2Reg 2026 PSMAReg challenge [9, 8, 11] formalises this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan, evaluated jointly on organ overlap, boundary distance, deformation regularity, and preservation of PET-derived tumour biomarkers.

Deformable registration has traditionally been posed as an iterative optimisation over a regularised similarity objective, with widely used instances including the diffeomorphic Demons [28], SyN [1], and B-spline [24] schemes. While accurate, these methods are slow at whole-body resolution. Learning-based methods instead amortise the optimisation over a training set into the network weights, replacing per-pair iterative optimisation with a single forward pass at inference. VoxelMorph established the unsupervised paradigm [3], and subsequent work introduced diffeomorphic parameterisations through stationary velocity fields [5], transformer backbones such as TransMorph [4], and multi-resolution architectures such as LapIRN [20]. Among the latter, LapIRN uses a Laplacian pyramid of coarse-to-fine sub-networks to recover large deformations while keeping the transformation smooth and invertible. A recurring limitation of purely intensity-driven losses is that they permit non-physical folding. Anatomy-aware regularisation that penalises non-rigid deformation, e.g. of skeletal structures [16, 23, 26, 19], is therefore particularly relevant for whole-body PET/CT, where the skeleton is a frequent site of PSMA-avid disease.

In this work, we adapt LapIRN to longitudinal whole-body PSMA PET/CT for the PSMAReg task.

Our contributions are:

- an adaptation of the diffeomorphic, coarse-to-fine LapIRN architecture to longitudinal whole-body PSMA PET/CT, combining an affine pre-registration with a three-level pyramid and test-time instance optimisation;
- a set of biomarker-preserving losses that constrain the PET-derived MTV and TLG under the estimated deformation, including a Jacobian-based surrogate that supplies dense gradients throughout the lesion interior;
- a per-structure rigid-fit bone-rigidity penalty
- a score-driven checkpoint and submission selection procedure that mirrors the challenge ranking.

The proposed registration pipeline with affine pre-registration, LapIRN-based registration, losses and instance optimization (IO) is shown in Fig. 1.

2 Methods

2.1 Data and Preprocessing

We used the PSMAReg dataset of the Learn2Reg 2026 challenge [14], a longitudinal whole-body PSMA PET/CT cohort adapted from the manually annotated autoPET data of [11]. Each paired subject provides a baseline scan and one or more follow-up scans, each with a CT and a PSMA PET channel; the follow-up is registered to the baseline, i.e. the baseline acts as the fixed and the follow-up as the moving image.

Of the 146 paired and 212 unpaired subjects in the challenge training split, we used the paired ones, divided patient-wise into 117 for training and 29 for internal validation. Registration pairs always point from the temporally later scan to the earlier one: for training we took all such ordered time-point combinations, including follow-up-to-follow-up pairs (239 pairs). For our internal validation subjects we used every follow-up registered to its baseline giving 39 pairs from the 29 subjects. The challenge validation split of 20 paired subjects served as our test set; the organisers' held-out test set ($\sim$200 pairs of private JHU data) is scored only by them.

For the training split we used the labels released by the organisers: CT organ labels generated automatically with TotalSegmentator [29] and PET lesion labels manually annotated by radiologists. Since no labels are released for the challenge validation split, we generated our own for our test set with TotalSegmentator for the CT organs and the winning autoPET III submission [22] for the PET lesions.

All images are provided already resampled to an isotropic in-plane spacing of $2.7344 \times 2.7344 \times 3.27$ mm and cropped to a fixed size of $192 \times 192 \times 288$ voxels, so no further geometric preprocessing was applied. We normalised CT by clipping to a fixed window of $[-1000, 1500]$ HU and PET by clipping the SUV to $[0, 20]$, each rescaled to $[0, 1]$.

2.2 Registration Pipeline

The pipeline consists of an affine pre-registration followed by a deformable LapIRN stage and, at test time, instance optimisation (Fig. 1). Throughout, I denotes the moving PET image, M its lesion mask, φ_{aff} the affine pre-registration and φ_θ the displacement field predicted by the network; their composition φ is the total transformation mapping the moving into the fixed image, and W the lesion region in the fixed frame, i.e. the image of M under φ.

Affine pre-registration and preprocessing. Baseline and follow-up scans of the same subject differ in field of view, table position and patient posture by far more than a locally regularised network can absorb, so we first remove the global component with a separate affine stage. We follow the baseline provided the challenger organisers, by removing the bed and normalizing the CTs. For each pair we estimate an affine transform with ANTs [2] on the CT channel alone, at half resolution and with a Mattes mutual-information metric, after masking both volumes to the body and windowing them to $[-1000, 1000]$ HU. The result is converted into a dense voxel displacement field, cached per pair, and applied to the moving scan before the network sees it. Note that an affine transform is in general not volume-preserving, which is why all biomarker terms of Sec. 2.3 are evaluated on the composed transform φ rather than on φ_θ alone.

Deformable registration. For the deformable stage we use LapIRN [20], a Laplacian-pyramid network whose three sub-networks operate at a quarter, a half and the full resolution of the $192 \times 192 \times 288$ volumes. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level; the sum is exponentiated by scaling and squaring to give a diffeomorphic φ_θ, so large deformations are recovered coarse-to-fine while the transformation stays smooth and invertible. We adapted the architecture to the multi-modal, whole-body setting: each sub-network takes four input channels, the CT and PSMA PET of the affinely aligned moving scan and of the fixed scan, and uses 7 initial feature channels with five residual blocks per level. Image similarity is computed on CT only, since PET uptake is not conserved between time points and would otherwise drive the deformation towards matching disease rather than anatomy. The PET channel enters the network as input and the biomarker terms of Sec. 2.3. The levels are trained in sequence, each initialised from the previous one, which is kept frozen for the first ten epochs and fine-tuned jointly afterwards.

2.3 Loss Functions

The full objective combines image similarity, weak label supervision, the PET-biomarker terms, the bone-rigidity penalty and two deformation regularisers:

$$\begin{aligned}\mathcal{L} = {} & \lambda_{\mathrm{NCC}}\,\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\,\mathcal{L}_{\mathrm{DSC}} + \lambda_{\mathrm{MTV}}\,\mathcal{L}_{\mathrm{MTV}} + \lambda_{\text{MTV-J}}\,\mathcal{L}_{\text{MTV-J}} \\ & + \lambda_{\mathrm{jac}}\,\mathcal{L}_{\mathrm{jac}} + \lambda_{\mathrm{TLG}}\,\mathcal{L}_{\mathrm{TLG}} + \lambda_{\mathrm{rigid}}\,\mathcal{L}_{\mathrm{rigid}} + \lambda_{\mathrm{smooth}}\,\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\,\mathcal{L}_{\mathrm{NDV}}.\end{aligned} \tag{1}$$

All terms are evaluated on the composed transformation φ. Only the full-resolution level is trained with the complete objective. The two coarser levels (quarter and half resolution) use the similarity, label and regularisation terms alone, i.e. $\lambda_{\mathrm{MTV}} = \lambda_{\text{MTV-J}} = \lambda_{\mathrm{jac}} = \lambda_{\mathrm{TLG}} = \lambda_{\mathrm{rigid}} = 0$, because the lesion and bone masks they would be evaluated on are only a few voxels wide there: after downsampling, thin ribs and small lesions are largely partial-volume, so both the volume ratios and the per-structure rigid fits measure interpolation rather than anatomy.

Registration Accuracy. Image similarity is measured by the local normalised cross correlation, evaluated as a multi-resolution sum over as many scales as the level has resolutions, i.e. one scale at the coarsest and three at the full-resolution level. It is computed on the CT channel only: PET intensity is not conserved between the two time points, as lesions appear, resolve or change uptake under therapy, so a PET similarity term would reward matching disease rather than anatomy. Furthermore, we used a standard DSC loss for weak CT label supervision. The objective contains no surface-distance term, although HD95 is scored; Sec. 3.2 shows that one would have nothing to improve.

Preservation of PET quantification. The scored MTV bias only constrains the *net* lesion volume, so we penalize it directly while supplying dense gradients through a Jacobian-based surrogate. All terms are evaluated on the composed transformation φ against the original moving lesion mask M, since the affine itself is generally not volume-preserving. The first term, $\mathcal{L}_{\mathrm{MTV}}$, is the squared relative error between the volume of the resampled mask and that of M. It matches the scored metric exactly, but its gradient is confined to the lesion's sub-voxel boundary shell. We therefore add a complementary term based on the change-of-variables identity: summing $\det J_\varphi$ over the warped lesion region W recovers the original lesion volume, so the mean determinant over W equals the ratio of original to warped volume. Driving that mean to one, $\mathcal{L}_{\text{MTV-J}}$ pins the same net volume ratio while providing gradients throughout the lesion interior, and unlike a per-voxel constraint it still permits internal redistribution of volume. Squaring the deviation per voxel instead of after the mean gives a third, stricter term, $\mathcal{L}_{\mathrm{jac}}$, which additionally discourages compensating compression and expansion inside the lesion.

The challenge additionally scores the total lesion glycolysis (TLG), the PET-activity-weighted counterpart of the MTV, which we penalize analogously as the relative change of intensity mass under the warp, i.e. the absolute difference between the sums of I over M before and after resampling, divided by the former. Since both the image and the mask are resampled with the same transformation, this term is sensitive not only to volume change but also to intensity blurring at the lesion boundary, and thus complements the purely geometric terms. All four are used during training and descended at test time by our instance optimisation (Sec. 2.4), so the network is trained on the objective it is later refined under.

Bone rigidity. A median of 46.0% of total lesion volume at baseline lay within skeletal structures, rising to 63.8% at follow-up. Since so much of the lesion volume is intraosseous, a rigidity penalty on a bone mask is an anatomically grounded surrogate for the biomarker terms: a rigid motion has $\det J_\varphi = 1$ and resamples the PET signal without compressing or expanding it, so it protects the lesion volume MTV measures and the intensity mass TLG measures at once.

We first used the established local rigidity penalty, which combines deviation of the Jacobian J_φ from orthogonality [16, 23] with local affineness and orientation preservation [26, 19]. Its terms are evaluated by finite differences at every voxel and only then masked, so the stencil also reads non-bone voxels and, on thin cortical bone and ribs, ends up measuring the bone-to-soft-tissue displacement discontinuity (Sec. 3.2); eroding the mask would fix this, but at whole-body resolution many bone labels are only a few voxels wide. We therefore replaced it with a per-structure rigid-fit residual over the 61 skeletal TotalSegmentator labels: for each structure we fit the best rigid transform to its own displacement in closed form by Kabsch alignment [12, 27] and penalise the mean squared residual, averaged within and then across structures. No stencil crosses the mask and each structure is fitted independently, so articulation is free, and unlike the derivative-based penalties [26, 19] it constrains only the shape of the deformation and not its placement.

Deformation regularisation. To regularise the displacement field we used a diffusion regulariser, $\mathcal{L}_{\text{smooth}}$, the mean squared spatial gradient of the displacement, together with a term penalising non-diffeomorphic voxels. Rather than the usual penalty on negative Jacobian determinants of a single finite-difference stencil, we differentiably reproduce the scored metric itself: the challenge computes the non-diffeomorphic volume by decomposing each voxel into six tetrahedra and accumulating the negative part of their determinants over the body mask [15]. Writing $\det J_\varphi^{(k)}$ for the determinant of the k-th tetrahedral scheme, the loss

$$\mathcal{L}_{\text{NDV}} = \frac{100}{12|\Omega|} \sum_{p\in\Omega} \sum_{k=1}^{6} \max\bigl(-\det J_\varphi^{(k)}(p),\, 0\bigr) \tag{2}$$

equals the scored percentage of non-diffeomorphic volume and places gradients on exactly the folded voxels, so that minimising it minimises the reported %NDV directly.

2.4 Instance Optimisation

At test time we refine each pair individually. Keeping the predicted field fixed, we optimise a residual stationary velocity field, initialised to zero and exponentiated by scaling and squaring, on top of it, so the refined transformation stays diffeomorphic by construction and the network is only ever corrected, never overwritten. We descend it for 100 Adam steps with a learning rate of 10^{-2} and keep the best rather than the last iterate, scoring each step with the DSC, HD05, MTV and TLG, so the selected step is the one the scorer would prefer.

Submission container. Instance optimisation needs CT organ segmentations of both scans and PET lesion segmentations of the moving scan, and the submitted container has to produce them and register the pair within a budget of 90 s per image pair, which the configuration above exceeds. The deployed method therefore differs from the one evaluated in validation in three points, all of them inside instance optimisation or in the labels it consumes; everything up to and including the predicted field is identical. First, PET lesion masks come from an nnU-Net [10] trained on this dataset with progressive patch-size growing [7], which is faster at inference than the ensemble used in validation. Second, CT organ labels are produced with TotalSegmentator in its fast mode together with its body segmentation. Third, instance optimisation runs until the 90s per-pair time limit is exhausted. We report both configurations separately in Sec. 3.1, since only the validation one is free of the time constraint.

2.5 Model Selection

Selection enters our pipeline twice, to keep a checkpoint within a run and to choose a variant across runs, and both use the challenge's own ranking score, the weighted geometric mean of registration accuracy, biomarker preservation and deformation regularity with exponents 0.4, 0.4 and 0.2. Selecting on accuracy alone would be wrong in either place, because the two do not peak together: validation Dice improves until it plateaus, whereas the tumour metrics reach their optimum roughly halfway through level-3 training and degrade thereafter (Fig. 2).

2.6 Baselines

We compare against two references and two competing methods, all evaluated on the same pairs and under the identical protocol of Sec. 3.1.

As a classical iterative baseline we ran NiftyReg [18, 17], using `reg_aladin` for the affine stage followed by `reg_f3d` initialised with it, both with default parameters. As a learning-free optimisation baseline we ran ConvexAdam [25], in the form provided by the challenge organisers as the official baseline of the task.

2.7 Implementation Details

Augmentation was chosen to reproduce the ways two sessions of the same patient differ rather than to maximise variety: a left–right flip ($p = 0.5$), a CT intensity shift of ± 0.02 in the normalised window (about ± 50 HU) and scaling of 0.9 to 1.1, and a PET scaling of 0.85 to 1.15 without a shift, since SUV depends on dose, uptake time and body weight but an additive offset has no physical counterpart. We additionally crop up to 40 slices from either end of both scans along the axial axis, and then the moving and fixed scan independently by up to 10 further slices each; the latter is what reproduces a baseline/follow-up mismatch, forcing the network to register anatomy present in one scan and truncated in the other.

Each pyramid level is trained in sequence with Adam, at a batch size of one with gradients accumulated over four steps, for 60k, 60k and 120k steps at learning rates of $3 \cdot 10^{-4}$, $2 \cdot 10^{-4}$ and $2.5 \cdot 10^{-4}$. Every level starts with a five-epoch linear learning-rate warmup and keeps the preceding level frozen for its first ten epochs. The convolutional trunk runs in `bfloat16` while the transformations and all losses are kept in single precision, since scaling and squaring and the Jacobian determinants compound rounding error. The loss weights were chosen by the sweep of Sec. 3.1.

All models were trained and evaluated on a single NVIDIA H100 80 GB HBM3 (driver 610.57.04), in Python 3.11 with PyTorch 2.6.0 [21] and CUDA 12.4;

the affine pre-registration uses ANTsPy 0.6.3 [2], and image and array handling NiBabel 5.4.2, NumPy 2.4.4 and SciPy 1.17.1. Training all three levels takes 2d 6h 7m 6s.

3 Results

3.1 Evaluation Protocol

The challenge does not release labels for the validation cases. Unless stated otherwise, the validation numbers we report are therefore computed against surrogate labels we generated ourselves: CT organ labels with TotalSegmentator and PET tumour labels with the winning autoPET-III submission. We report our method in three configurations: *ours (container)*, the submitted method of Sec. 2.4, which is what the hidden test set will be scored on and therefore the one we test the baselines against; *ours (validation)*, the unconstrained method of Sec. 2.2, reported as the upper bound the 90 s per-pair budget gives up; and *ours (no IO)*, the backbone alone, which isolates what test-time refinement contributes. The latter two carry no significance marks.

3.2 Ablation of the Objective

Derivative-based rigidity is ill posed on thin bone. We quantified the stencil leakage of Sec. 2.3 across five cases: 53.0% (range 51.6 to 54.7) of the voxels entering the loss were not bone, and 44.8% of the gradient magnitude fell outside bone. A field with zero displacement inside every bone, which is perfectly rigid by construction and should therefore cost nothing, scored 56 (range 40 to 69), all of which vanished after a single mask erosion. The term was thus measuring the bone-to-soft tissue displacement discontinuity that any correct registration must exhibit. With the per-structure rigid-fit penalty of Sec. 2.3, the zero displacement field now scores exactly 0 and a per-bone rigid field 2.4e-6, while anisotropic squeeze is still rejected at 1.16.

A surface-distance loss would not help. Since HD95 is scored, a surface-distance term [13] is the obvious addition to the objective, but the 2296 scored (case, label) entries leave it nothing to do. Half the HD95 mass sits in the worst 10% of entries, and splitting it against the unregistered pair shows that 48.1% is inherent, already bad before registration and no better after, while none of it was created by our field. Those entries are limbs repositioned between sessions, hollow organs with changing filling and anatomy truncated by the field of view, none of which have a boundary to refine. The rest is already at the discretisation floor: the median per-label HD95 is one voxel and the third quartile two. We measured this before instance optimisation, so it holds for the training objective as well.

Table 1. Our method against the baselines on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. 3.1. Every cell is a mean over the cases, runtime included: DSC and the MTV/TLG errors in %, NDV in ppm, and time in seconds per image pair on the hardware of Sec. 2.7. Bold marks the best value in each column. * marks metrics on which *ours (container)* is significantly better than the baseline in that row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$); runtime is not tested. Dashes mark metrics undefined for a baseline.

Method	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓	Time (s) ↓
Initial	14.8*	30.9*	–	–	–	–
Affine	52.9*	11.7*	5.47	4.70	–	**19.1 ± 0.920**
ConvexAdam	63.0*	8.41*	5.10	4.15	100	22.8 ± 1.06
NiftyReg	73.3*	7.09*	4.46	8.68*	1525*	49.7 ± 0.480
Ours (no IO)	76.5	6.65	3.22	5.08	**0.00**	19.5 ± 1.02
Ours (container)	81.0	6.17	3.29	3.31	864	89.8 ± 2.69
Ours (validation)	**88.3**	**4.80**	**1.37**	**2.13**	70.9	383 ± 7.77

3.3 Comparison against Baselines

We swept the loss weights and the instance-optimisation settings and selected the one reported below with the challenge ranking of Sec. 2.5, applied to the surrogate protocol above. Table 1 places the three configurations against the baselines. The container configuration improves DSC from 73.3% (NiftyReg, the strongest baseline) to 81.0% and reduces HD95 from 7.09 to 6.17 mm, while also reducing the MTV and TLG errors and the non-diffeomorphic volume from 1525 to 864 ppm.

3.4 Leaderboard Performance

On the validation leaderboard, our method (*Ours (validation)*) achieved a Dice score of 0.789 ± 0.0381, HD95 of 5.99 ± 2.39, MTV of 1.56 ± 1.22, TLG of 4.18 ± 2.49 and NDV of 0.0041 % ± 0.0024%.

4 Discussion

The biomarker terms are cheap to add and are descended identically during training and at test time, so the network is refined under the objective it was trained on - they help reduce the MTV and TLG bias against every baseline without costing accuracy.

Derivative-based rigidity penalties are not wrong in general, but they assume that a structure is thicker than the stencil used to differentiate it. On whole-body

CT at this resolution, ribs and cortical bone are not, so the term reads mostly soft tissue and measures displacement discontinuity that a correct registration must produce. The per-structure rigid fit avoids this by reading nothing outside each label, and we would expect it to matter wherever rigid structures are thin relative to the voxel size, rather than as a universal replacement.

Several limitations qualify these numbers. All validation metrics are computed against automatic surrogate labels, which carry their own error; the model was selected on the same cases it is evaluated on, so the comparison against the baselines is exploratory rather than confirmatory; and no test-set results were available at the time of writing. The container configuration also loses accuracy due to its 90 s per-pair budget, most of it in the segmentations that instance optimisation consumes, so a faster and more accurate segmentation stage is the most direct improvement available.

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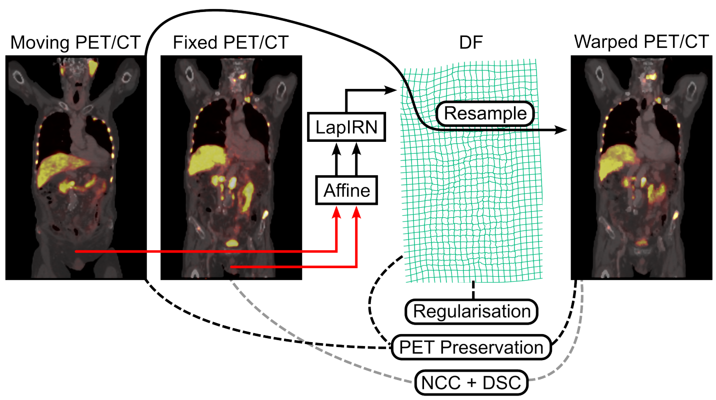

Fig. 1. Overview of the proposed registration pipeline. The moving (follow-up) and fixed (baseline) PET/CT are first coarsely aligned by an affine pre-registration (red), and the affinely aligned pair is passed to the coarse-to-fine LapIRN network, which predicts a dense displacement field (DF, green grid) used to resample the moving scan into the fixed space (warped PET/CT). Training is driven by three groups of objectives: deformation regularisation on the DF, comprising a global smoothness and folding penalty and a per-structure bone-rigidity term that constrains the DF inside skeletal labels and thereby also acts as an anatomical surrogate for biomarker preservation, since a large fraction of the lesion volume is intraosseous; PET-biomarker preservation terms (MTV, TLG and Jacobian inside the tumour mask) comparing the moving and warped lesions (black dashed); and image- and label-based similarity, NCC and DSC, between the fixed and warped scans (grey dashed). At test time, the same objectives are descended per pair by instance optimisation (IO).

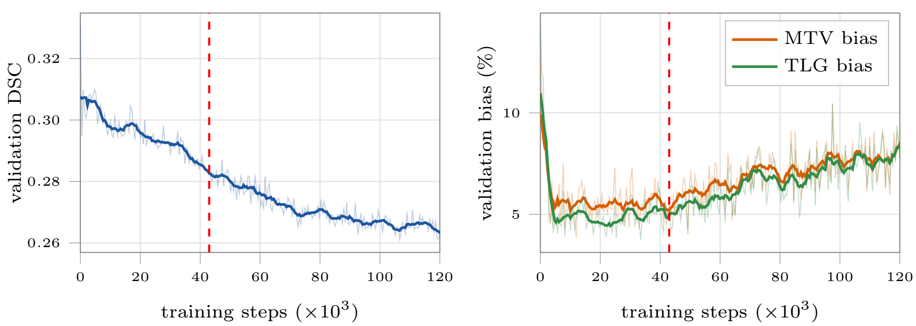

Fig. 2. Accuracy and biomarker preservation do not peak at the same checkpoint. Validation CT Dice (left) rises to a plateau, while the MTV and TLG bias (right) reach their optimum roughly halfway through level-3 training and degrade thereafter. The red dashed line marks the checkpoint selected by Eq. (**??**). Faint lines are the per-round values, solid lines a centred rolling mean.